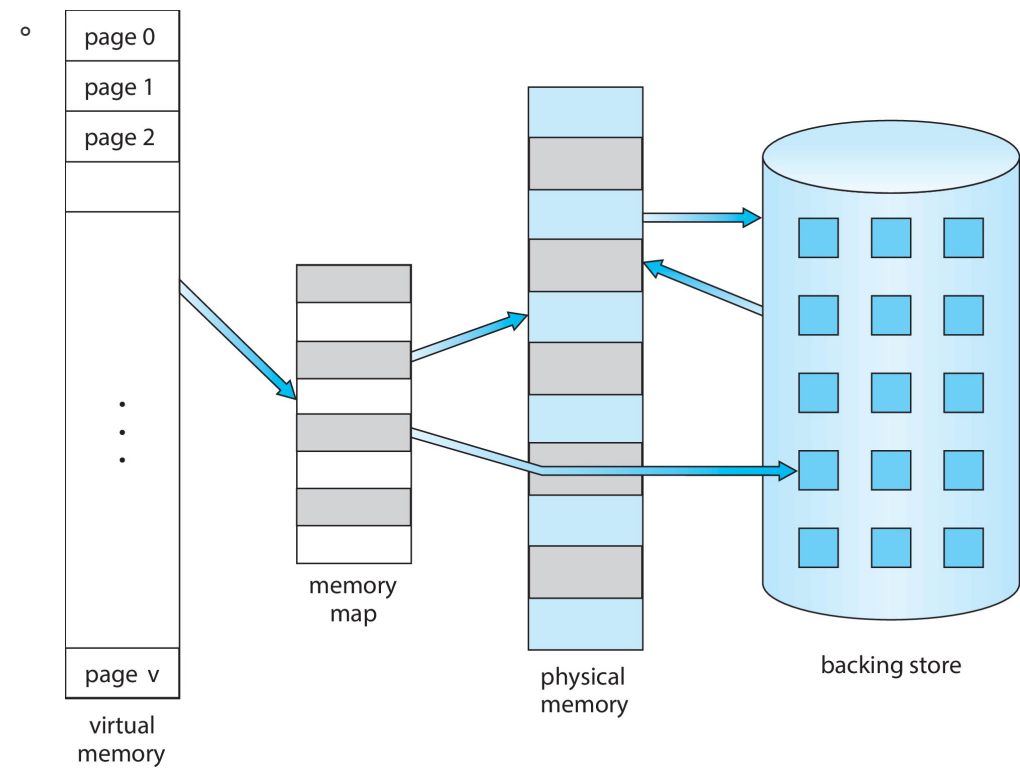


Chap9_Virtual Memory

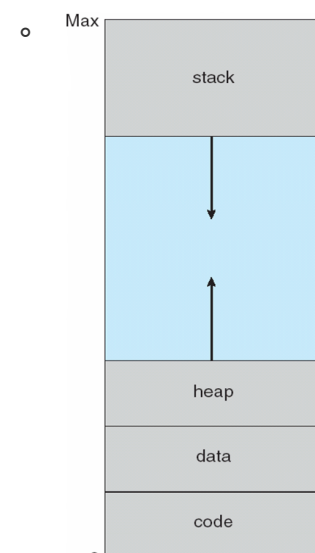
- Introduction

- Virtual memory is a virtual table which has higher memory than the actual physical memory
- Because all part of codes are not required to be in memory all the time,
- Less I/O needed to load or swap programs into memory
- **Virtual addr space** - logical view of how processes are stored in memory
- Can be implemented via
 - Demand Paging
 - Demand Segmentation

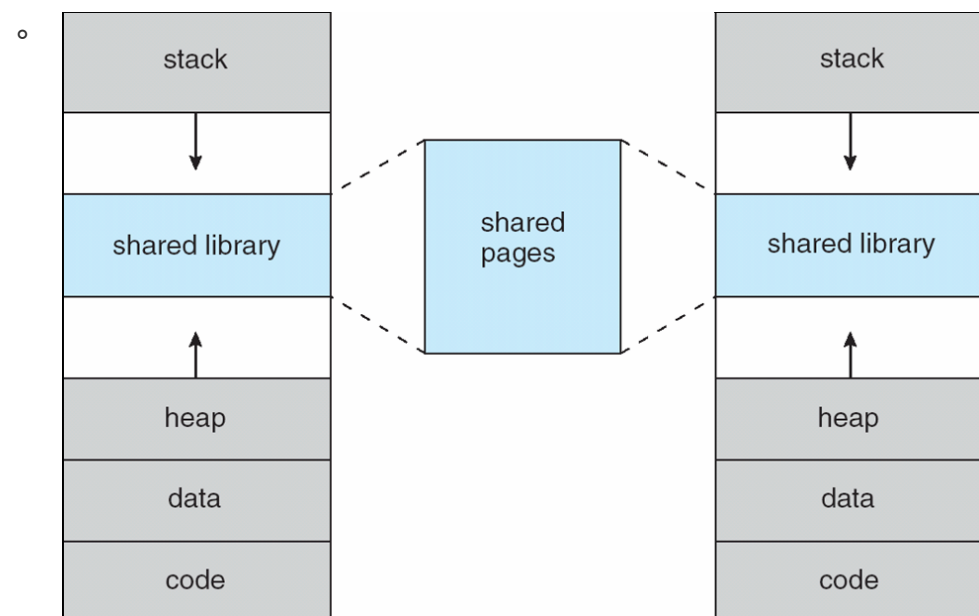


- Virtual address Space

- Stack starts from top(max) and grows down while the heap space grows up

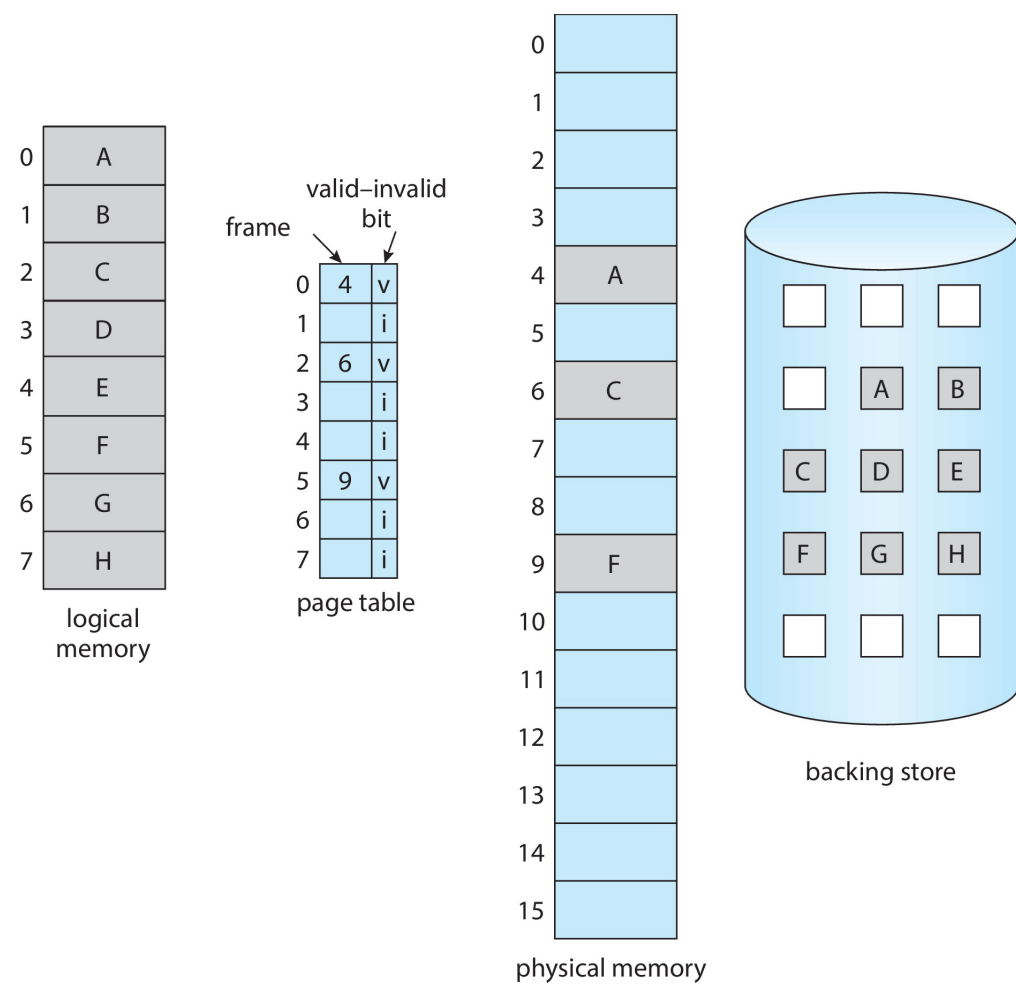


- The virtual space left in between is for the shared memory and system libraries.



• Demand Paging

- Retrieve page only when required.



- Lazy Swapper. Swapper that deals with pages is a pager
- **Similar to apging system with swapping (a backing store).**
- When page is needed, reference to it
 - i - invalid ref. - abort
 - not in memorv - bring to memorv

- v - in memory

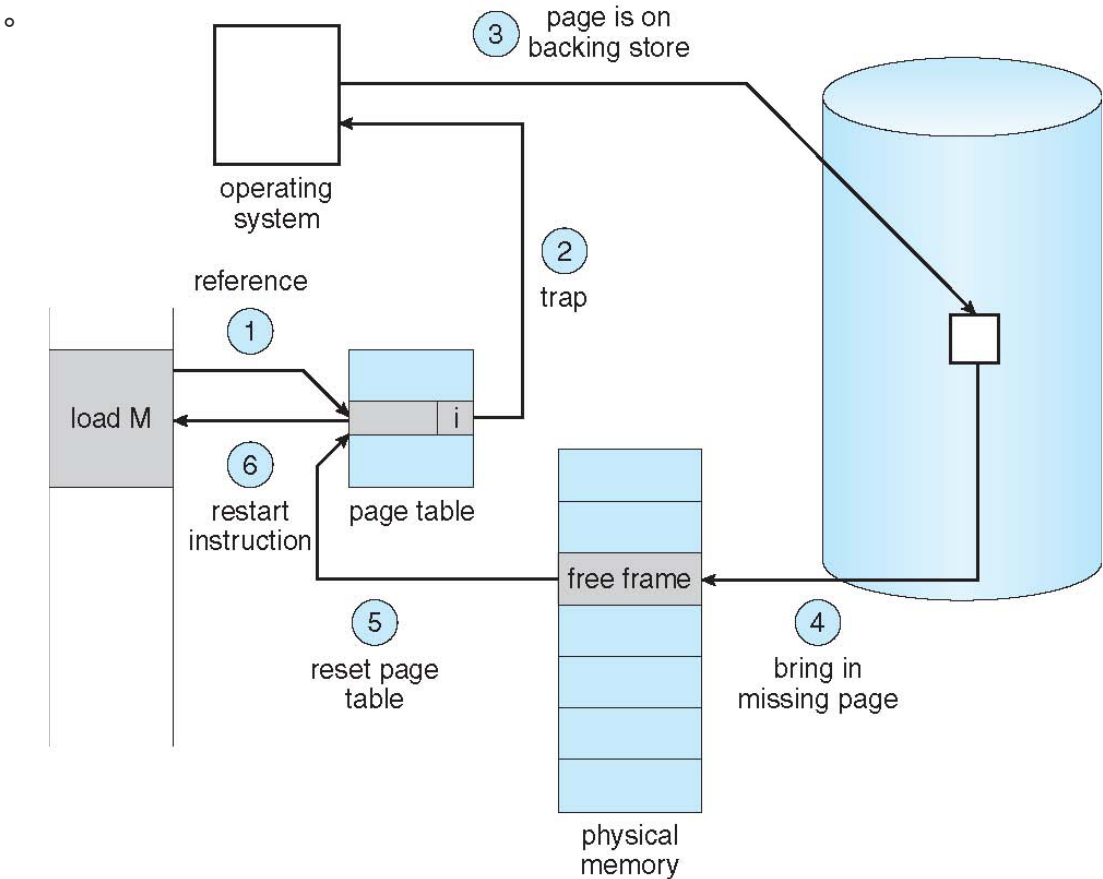
Frame #

valid-invalid bit

	v
	v
	v
	i
...	
	i
	i

page table

Pages to be removed from physical memory are guessed based on algorithms such as LRU LFU MFU etc. **Done by MMU.**



- Aspects of Demand Paging

- Swapping backing store hardware is required
- Starting with 0 pages in memory could result in page fault for every process.**Pure demand Paging**
- **Page table with valid/invalid bit**

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